Total no. of Pages: 2 Roll no. .....

Degree: B.Tech. Semester: 3 Course work

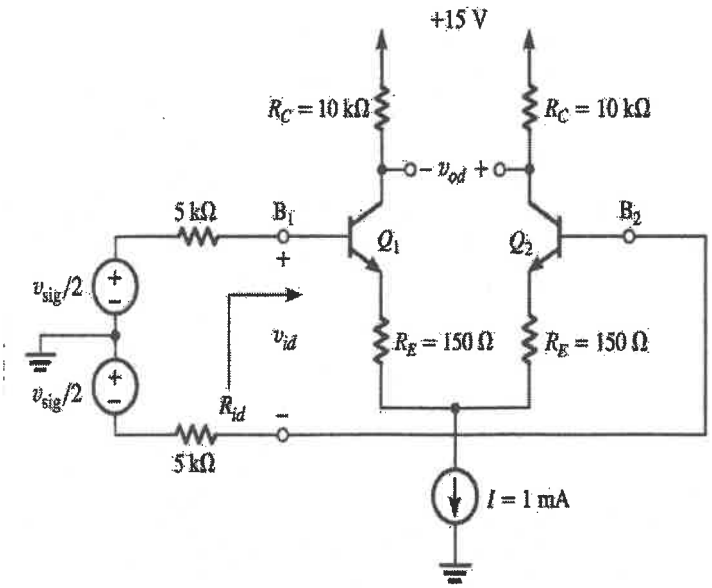
MID-SEMESTER EXAMINATION, SEP 2026

Course Title: Microelectronics Circuits and Applications

Course Code: ECECC304

Duration: 1:30 Hours Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

	Questions	**Marks**	**CO**
1a	Derive the expression for the overall voltage gain of an emitter follower and state the need for using emitter follower circuits.	**2**	**CO1**
1b	A BJT differential pair is biased with a current source I_{EE} = 1 mA. The collector resistors are R_C = 10 KΩ with a 1% mismatch (ΔR_C = 100 Ω). The output resistance of the bias current source is R_{SS} = 200kΩ. Assume V_T = 25 mV and ro = ∞. Calculate: i. The differential voltage gain (A_d) ii. The common-mode voltage gain (A_{cm}) for a single-ended output iii. The Common-Mode Rejection Ratio (CMRR) in decibels (dB)	**2**	**CO1**
2a	Draw the circuit for MOS differential amplifier and drive the expression for differential and common mode voltage gain.	**2**	**CO1**
2b	For the differential amplifier shown in Figure below with β = 100, determine the input differential resistance R_{id} and the overall differential voltage gain v_{od}/v_{sig}, neglecting the effect of r_o.  	**2**	**CO1**

3a	Draw the small-signal equivalent circuit of a Darlington pair and derive the expressions for its total current gain (A_i) and equivalent input resistance.	**2**	**CO2**
3b	A common-emitter amplifier has coupling capacitors C_{C1} and C_{C2} and an emitter bypass capacitor C_E. The corresponding resistances seen by these capacitors are R_{C1}, R_E, and R_{C2}, respectively. Using the short-circuit time-constant method, derive τ_{C1}, τ_{CE}, and τ_{C2}, and hence determine the lower 3-dB frequency f_L.	**2**	**CO2**
4a	For a cascode amplifier, derive the expression for the 3-dB frequency f_H using the open-circuit time-constants method. Explain how the choice of load resistance R_L affects the Miller effect and the high-frequency bandwidth when R_{sig} is large.	**2**	**CO2**
4b	A MOS cascode amplifier consists of two identical NMOS transistors (M1 and M2) operating in saturation. Bias parameters: I_D = 0.5 mA, g_m = 2 mA/V, r_o = 50 kΩ. The load is an ideal active current source ($R_L = \infty$). Calculate the overall output resistance (R_{out}) and intrinsic voltage gain ($A_v = V_o/V_{in}$) of the cascode stage.	**2**	**CO2**
5a	Explain the internal capacitances of MOSFETs for high-frequency operation, and also draw the equivalent small-signal model of MOSFET for high frequency.	**2**	**CO3**
5b	A source follower operated at g_m = 2 mA/V and r_o = 20 kΩ is fed by a signal source with R_{sig} = 10 kΩ and loaded by R_L = 20 kΩ. The MOSFET has C_{gs} = 20 fF, C_{gd} = 5 fF, and g_{mb} = 0.2gm. The total capacitance at the output is C_L = 15 fF. Given Q = 0.42, determine A_M, f_T, f_Z, f_{P1}, f_{P2}, and f_{3dB}.	**2**	**CO3**